

Finite Element Modeling and Design of a Pneumatic Braided Muscle Actuator with Multifunctional Capabilities

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Abstract—This paper reports the concept, design and development of a pneumatic braided muscle actuator, able to produce bi-directional force and motion. Such a capability can simplify the design and enhance the capabilities of robotic/automation systems using pneumatic muscles as actuators. Finite element modeling is utilized as a design tool in order to study the feasibility of the concept, where a single braid structure is deformed to produce both contraction and elongation. A 3D-finite element model of the actuator is developed and a number of non-linear quasi-static simulations are performed using the explicit dynamic solver LS-Dyna®, to measure the blocked force and free displacement of the actuator. The effect of varying the end-fittings diameter and the length of the inner chamber, on the mechanical characteristics of the actuator is also analyzed. The simulation results are validated through mechanical experimentation performed on the functional prototype. With an actuation pressure of 1 bar, the actuator is able to produce a contraction force of around 80 N, extension force of around 40 N and an overall stroke of >40% of the total length of the actuator. Moreover, the novel actuator has been utilized as main driver for a one degree of freedom variable stiffness joint, as a case study, to demonstrate the usability of the actuator.

Index Terms— Finite element modeling, Soft actuators, Pneumatic braided muscles, Flexible fluidic actuators, Variable stiffness.

I. INTRODUCTION

PNEUMATIC Artificial Muscles (PAM), more commonly known as fluidic or Mc-Kibben muscles have been around for quite some time now. They are generally good actuation candidates when high power to weight ratio, lightweight, ease

of installation, hazard free use and inherent compliance are required. These properties make them an ideal choice for industrial/robotic applications where human-machine interaction is involved. Recent and comprehensive surveys on their applications can be found in [1][2]. PAM usually consists of a hollow cylindrical elastomeric chamber covered by an outer braided sleeve, consisting of fibers made of un-stretchable material and arranged in an anti-symmetric helical configuration. The hollow internal chamber and the braided sleeve are tightly sealed and attached to rigid end-fittings, usually a passage is provided through one of these end-fittings for pressurizing the elastic chamber with air or another fluid. The mechanical work done is transferred to an external system through these end-fittings. When the inner elastic tube is pressurized the muscle either expands or contracts or stiffens depending on the initial angle of the braid fibers with the longitudinal axis of the muscle. The existing/traditional PAMs have a fixed initial braid fiber angle and are usually designed to produce a contractile force upon actuation, hence they are able to produce uni-directional force and motion [3]. This single acting nature of traditional PAMs poses a drawback when utilized for robotic applications (e.g. for bi-directional actuation of a robotic joint) since it is required to provide two actuators in antagonistic configuration. This increases the overall size and complexity of the actuation mechanism [4]. In order to overcome this challenge, previous attempts have been made to develop bi-directional PAMs.

In [5] a spring over muscle (SOM) actuator is reported. The SOM utilizes a passive compression spring in parallel with a PAM for producing bi-directional forces. Zheng & Shen developed a double acting sleeve muscle actuator [6], which is able to produce bi-directional force and motion. The muscle incorporates a unique insert at the center, inside the hollow elastic chamber. This design results in a significant decrease in the air consumption during operation and an increase of force capacity over the entire range of contraction. Furthermore, the insert consists of an additional chamber, which can produce extension, when pressurized. Although the fluidic muscle is able to produce bi-directional force and motion, it has some disadvantages. First, the insert increases the weight of the actuator; secondly, the extension force and motion is quite low as compared to contraction force and motion of the fluidic

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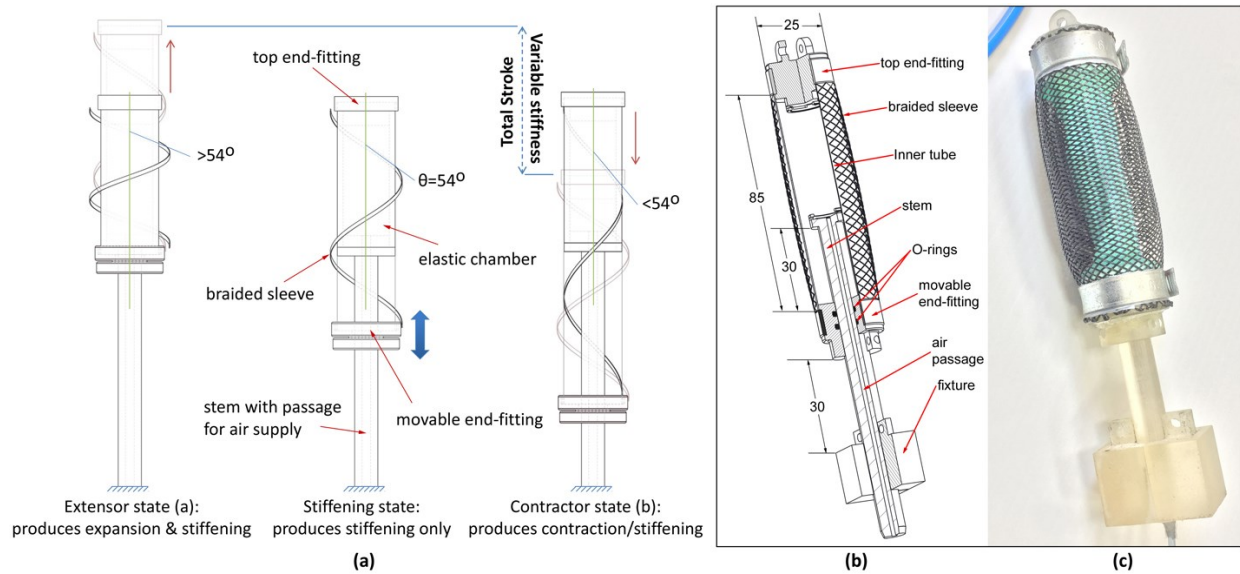


Fig. 1. (a) Actuator working concept, showing the deformation of the braided sleeve for achieving the extensor and contractor states of the muscle. A single braid fiber is shown instead of the whole outer braided sleeve for clarity. (b) Sectioned CAD rendering of the muscle. (c) Physical prototype. Dimensions in mm.

muscle. Moreover the lateral flexibility of the actuator is compromised as a rigid insert is occupying the inner chamber. The patent in [7] discloses a pneumatic actuator, which is able to achieve dual actuation states, that is able to contract and expand upon actuation, but it achieves this by circulating the working fluid between two expandable fluid-containing cells. The concept of inverse pneumatic artificial muscle (IPAM) is evaluated in [8], where a transition between low pressure and high pressure inputs is used to produce large strains (300%) and forces.

Different approaches have been used previously to model PAMs with the aim to relate the input pressure and its length to the force it is able to produce along its entire axis. An analytical model based on the principle of virtual works is presented in [9], that relates the different attributes according to the following equation;

$$F(\varepsilon, P) = (\pi r_0^2) P [a(1 - \varepsilon)^2 - b] \quad (1)$$

$$\varepsilon = (l_0 - l) / l_0, a = 3 / \tan^2(\alpha_0), b = 1 / \sin^2(\alpha_0)$$

Where F is the force produced, P is the applied pressure, r_0 is the initial radius, l_0 is the initial length, l is the final length while α_0 is the initial braid angle. A good survey on the different mathematical models can be found in [11][12]. Another popular approach to model the highly non-linear behavior of PAMs is by using Finite Element (FE) modeling tools. These are usually preferred for their capability to simulate hyper-elastic materials and their intuitive visual representation of the results. Antonelli et al. propose a non-linear FE model in [13], which takes into account the non-linearity of the rubber material for the inner chamber using 8-node brick elements. The braid structure is modeled as rhombus-shaped elements, using 2-node trusses, with each side presenting a portion of the strand of threads. The external trusses nodes and the inner brick elements nodes are always

connected and no friction is considered among the strands. Furthermore, the 3D model only simulates one-eighth of the physical prototype by assuming symmetry on the plane perpendicular to muscle axis and due to axis-symmetric geometry of the muscle. A FE model for a pneumatic bending actuator is proposed in [14]. The inner tube is modeled as solid elements with hyper-elastic material properties while the external braid is modeled as helical threads patterning around the longitudinal principle axis of the inner tube by using quadratic beam elements. Similar to [13] the fiber braiding and the rubber tube are permanently attached, so there is no sliding motion between these entities. The bending is achieved by utilizing different braid structures (with different braiding angle for the fibers) on the two halves of the cylindrical actuator. Upon actuation, the half with the angle greater than 54° tends to expand while the half with the angle less than 54° tends to contract, hence achieving bending. Takashi and Toshiro [15] use 3D FE-modeling to establish a quantitative design optimization method for Mc-Kibben type PAMs. Similarly, the influence of fiber angle on the deformation of fiber-reinforced soft fluidic actuators is investigated through FE modeling in [16].

In this paper, we present the concept, design and the development of a “low pressure” pneumatic braided muscle actuator, shown in Fig. 1, with multifunctional capabilities including bi-directional force/motion and the ability to achieve maximum stiffness at each point along the total stroke of the actuator (a capability lacking in traditional Mc-Kibben type actuators). “Low-pressure” because low input pressure value of 1 bar is used for the operation of the physical prototype. Finite element (FE) modeling is utilized as a design tool to check the practicability of the concept. The system is able to produce a maximum force of around 80 N, making it suitable for application in certain robotics fields’ e.g. Biomedical robotics, rehabilitation etc. The use of low input pressure (1-2 bar) in comparison to high pressures (5-10 bar) increases the

safety factor for the user in case of an accident/failure but in principle the use of higher pressures can produce much higher forces required for automation/ industrial robotics. The effect of varying the end-fitting diameter and the length of the inner chamber on the mechanical characteristics of the actuator is also analyzed. In section II the working concept of the actuator is explained followed by the development of the FE model reported in section III. The design of the muscle and its experimental characterization are presented in section IV and V, respectively. The results are reported and discussed in section VI, while section VII briefly describes a case study about a one degree of freedom (DOF) variable stiffness joint that uses the bi-directional muscle as its main driver. Finally, section VIII concludes the paper and report some future further improvements.

II. WORKING CONCEPT OF THE ACTUATOR

In PAMs, when the inner elastic chamber is pressurized, it tends to expand, also expanding the braided sleeve with it. It is known from literature that when the braided sleeve expands, its individual fibers reorient themselves in such a manner that it tends to attain an angle of around 54° , with the longitudinal axis. If the braid fiber angle θ is less than 54° , upon pressurization there is an increase in the radial direction and the muscle contracts. If the braid fiber angle θ is greater than 54° , upon actuation, there is a reduction in radial direction and the muscle tends to expand [10]. Finally, for θ equal to 54° , the braid structure already has its maximum volume and in case of further pressurization, the muscle tends to stiffen only with no additional deformation.

The working concept of the multifunctional pneumatic muscle is based on the traditional Mc-Kibben muscles, but some fundamental changes to the design enable the user to change the initial braid fiber angle θ independently, hence achieving bi-directional motion and forces upon actuation, as shown in Fig. 1(a). The braid structure is deformed axially first, by positioning the movable end fitting and then actuated by pressurizing with air. The ability to change the initial braid fiber angle enables the actuator to achieve maximum stiffness at each point along the total stroke of the actuator. Traditional PAMs lack this ability and are able to achieve maximum stiffness only at maximum displacement. In the extensor state, the muscle is able to produce extension upon actuation, similarly in the contractor state the muscle is able to contract. In the stiffening state, the actuation increases only the stiffness of the actuator, with no visible deformation.

III. FINITE ELEMENT MODELING AND SIMULATION

The following section reports the development of the 3D-FE model that can be effectively used to predict the behavior of the bi-directional muscle. A number of nonlinear finite-element simulations were performed using the explicit dynamic solver Ansys/LS-Dyna® [17] to measure the blocked force and free displacement in relation to the actuation pressure. LS-Dyna is a commercial explicit dynamic solver, particularly well-suited for solving highly nonlinear analysis including complex contact interfaces and models with non-

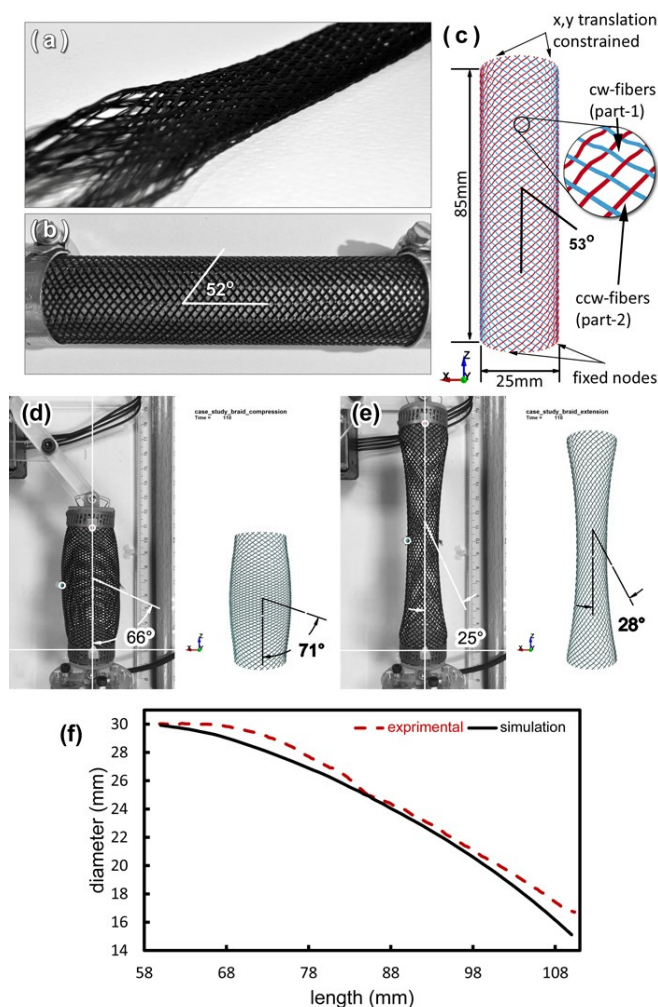


Fig. 2. (a) Commercially available braided sleeve. (b) Part of the braided sleeve annealed over a steel cylinder with a diameter of 25 mm. (c) FE model for the braided sleeve. (d)(e) Physical prototype and FE models showing the deformation of the braided sleeve. (f) Comparison between simulation results and experimental results for braid deformation.

TABLE I
BRAIDED SLEEVE PROPERTIES

	Braid length	Braid diameter	Fiber diameter	Fiber angle	No. of fibers	Fiber material
Physical	85	25	0.25	52°	112	PET
FE model	85	25	0.29	53°	56	$\rho = 1.35e^{-6}$ $E = 6$ $\nu = 0.37$

Units: mm, kg/mm³, Gpa.

TABLE II
MUSCLE MATERIAL PROPERTIES

	DENSITY, ρ	Young's Modulus, E	Poisson's Ratio, ν	Mooney Rivlin Constants
Braid	$1.35e^{-6}$	6	0.37	---
Inner chamber	$9.8e^{-7}$	---	0.495	$C1=1.809e^{-4}$ $C2=1.809e^{-5}$
End-fittings	$7.8e^{-6}$	210	0.29	---

Units: mm, kg/mm³, Gpa.

linear material properties. Such capabilities makes it an ideal choice for analyzing the nonlinear behavior of braided pneumatic muscles. The braid structure plays an important role in the operation of the actuator therefore an attempt was made to model the braid structure as realistically as possible. Previous studies tend to simplify the model of the braid by presenting it as a combination of rhombuses connected

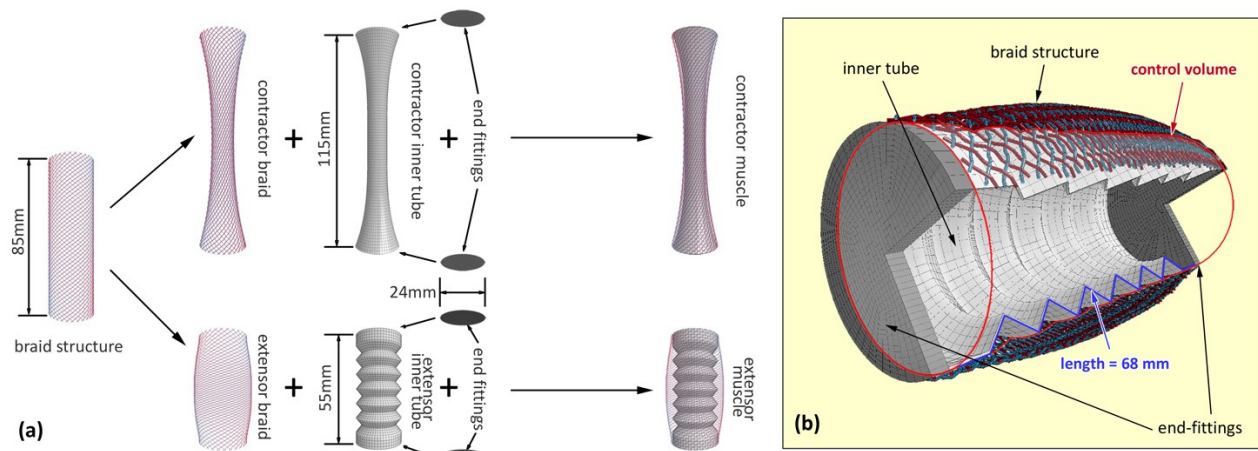


Fig. 3. FE model of the muscle. (a) Assembly of the muscle model, including the braid structure, the inner tubes and two end-fittings, for both states. (b) Detailed section view of the extensor state model. The red outline denotes the control volume; the space enclosed by the inner surfaces of the inner tube and both end-fittings.

together, forming a cylindrical structure, ignoring the overlapping of the fibers. First, a brief study of the braid structure is presented where a single braid structure is extended/contracted axially and the deformation in the structure is compared with the physical counterpart. Then, a similar process is used for producing the two states of the braided sleeve for the muscle models, followed by a detailed description of the development of the muscle model and the simulations performed.

A. FE Model of the Braid structure

The aim of this first step was to model a section of the commercially available braid structure and to study its axial/radial deformation and change in braid angle. These commercial braid structures are generally made of nylon or Polyethylene-Terephthalate (PET), and usually used for manufacturing Mc-Kibben type pneumatic muscles. The commercially available braid structure is shown in Fig. 2(a), the same piece of the braid is shown annealed over a steel cylinder in Fig. 2(b). Making a 3D geometry of the interlaced braided sleeve can be a tedious and time-consuming job, thus a single braid structure was generated using an open source software called "Pyformex" [18]. The software uses python script in order to generate complex geometrical models of 3D-structures. The geometry was then imported into LS-Prepost, where it was cleaned and re-meshed into 8,400 beam elements (3-node; element type 1 Hughes-Liu beam with cross-section integration). The beam has a circular cross section with diameter equal to 0.29 mm. The dimensions of the physical braided sleeve and the FE model are given in Table I. The braid was modeled as two parts. One part contains the clockwise (CW) fibers and the second comprises of the counter-clockwise (CCW) fibers (Fig. 2(c)). An isotropic elastic material model was used for the fibers, with material properties for PET, taken from literature [19] (Density, $\rho=1.35\text{E-}6 \text{ kg/mm}^3$, Young's modulus, $E=6 \text{ GPa}$, Poisson's ratio, $\nu=0.37$). Both parts share no-common nodes. An automatic general beam-to-beam contact is applied on the model with a static friction coefficient of 0.15. The

bottommost nodes of the CW and CCW fibers are fixed in space, while the topmost nodes are constrained and only Z-direction translation is permitted. A boundary prescribed motion of $\pm 25 \text{ mm}$ is applied to topmost nodes of the braided sleeve. The distance is covered in 50 ms and then kept at the deformed state for further 50 ms. The simulation time was kept short compared to the actual physical experiments for the deformation of the braid, in order to reduce the computational time. The Hughes-Liu beam model takes the cross section of the beam into consideration during stress calculations. Moreover two-nodes defines the body, while the third node is used to determine the orientation of the beam. The cross section integration and the orientation are required for the working of the automatic general beam-to-beam contact algorithm. It was noted that the static friction coefficient improved the stability of the simulations. The boundary conditions applied, constrain the braid model in a similar manner to the physical counterpart. The kinetic energies were monitored throughout the analysis to ensure that it was negligible compared to the total internal energy, as recommended by LS-Dyna developers [20] for quasi-static simulations. For the physical experiments a setup was designed to convert the rotational motion of a smart servomotor (Robotis, Dynamixel) into translation motion, used to displace one end of the physical braid while the other end is fixed to a load cell (ATI® Mini45-E Transducer). The deformation in the physical braid structure is measured from high definition videos analyzed with an open source software called Tracker® [21].

The results of the simulations and the experiments are presented in Fig. 2(d-f), showing good correspondence. The FE model can be effectively used to predict the deformation of the braid structure. The model and the physical braid is similarly deformed structurally, moreover the diameter/ length graphs follow a similar profile with overlapping points. The error can be attributed due to the minor differences in the model and the physical counterpart, especially the difference in the number of individual fibers. For both the simulations

and experimentation, the deformation force required was almost negligible (<1 N). Similar procedure is used for producing the deformed braided sleeve for the two states of the muscle.

B. FE Model of the Muscle

The non-linear behavior of the muscle is due to the large structural deformation of the braided sleeve and the non-linear material behavior of the inner tube. The FE model of the braid structure, explained in the previous section, was used to produce the deformed braided sleeve for the two states of the muscle. For the extensor state, the braid was compressed, while for the contractor state the braid was extended up to 30 mm. Two separate meshes were generated from these simulations. The initial stresses developed by the structure during deformation are not considered during the simulation of the muscle. Two models are developed to represent the extensor and the contractor states of the muscle. The inner tube is modeled using 4-node fully integrated shell elements with a thickness of 0.25 mm. The extensor inner tube has a length of 68 mm and an outer diameter of 24 mm, discretized by 56 elements along its modeled length and 56 elements along the circumference. Moreover, the chamber has a bellows-like profile, in order to accommodate the 68 mm length with in the extensor braid, which has a length of 55 mm. For the contractor model, the inner tube was modeled as a stretched cylindrical tube, having a non-uniform diameter, with a maximum value of 24 mm at the end-fittings and a minimum value of 11 mm at the center and a total length of 115 mm. The mesh consists of a total of 4480 elements. A hyper elastic material model (two-parameter Mooney-Rivlin rubber) is used for the inner tubes with properties of thin latex [22] (Table II). A clearance of at least 0.5 mm is kept between the inner tube and the braid structure to insure that there is no initial penetration of the beams into the shell elements. The automatic beam to surface contact algorithm is applied between the beams of the braid structure and the shell elements of the inner tube. A static friction of 0.15 is used. For the inner tube, it is assumed that there are no initial stresses present. The end-fittings are modeled as 4-node shell elements with a thickness of 2 mm. An elastic material model is used for the end-fittings with properties to approximate steel [18] (Table II). The assembly of both states of the muscle is shown in Fig. 3(a), together with a detailed sectioned view of the extensor state muscle, shown in Fig. 3(b). The extensor and contractor models of the muscle consist of 13,048 elements (beam=8,400 - shell=4,992) and 14,728 elements (beam=8,400 - shell=6,328) respectively.

The topmost nodes of the braided sleeve, the inner tube and the outermost nodes of the top end-fitting were constrained from moving relatively to each other. Similar constraints were applied on the bottommost nodes of the braid and tube plus the outermost nodes of the bottom end-fitting. The pressure load is applied to a control volume, enclosed by the inner-surfaces of the tube and the end-fittings. For the blocked force measurement, both the top and bottom end-fittings are fixed in space. A pressure load is applied, starting from zero and

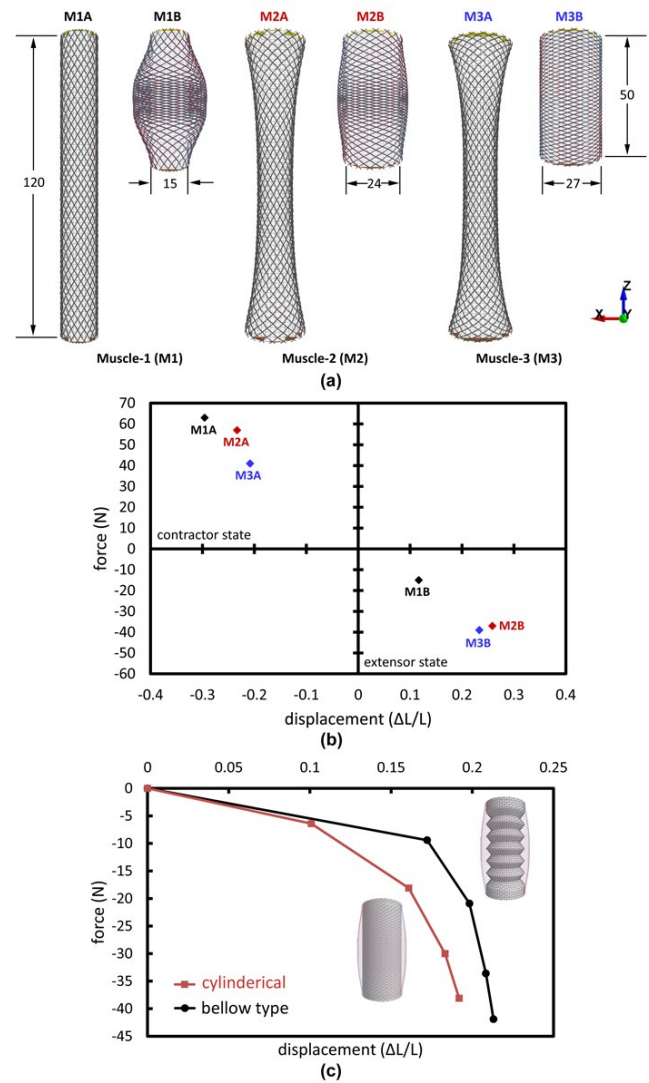


Fig. 4. (a) FE models of the simplified muscle with varying end fitting diameters. State A represents the contractor state while state B represents the extensor state. Dimensions in mm. (b) Graph showing the force capacity vs. free displacement for the different muscle models with varying end-fitting diameters. Muscle-2 (M2) is able to produce good forces and displacement in both states. (c) Graph showing the force capacity vs. free displacement for the cylindrical type inner chamber and bellows type inner chamber. Actuation pressures (0, 0.2, 0.5, 0.8, 1 bar).

reaching a certain value in 250 ms and then kept constant. Similar response time is common for the pneumatic braided actuators. The total simulation time for the experiment is 300 ms. For the free displacement measurement, the bottom end-fitting is fixed in space while the top end-fitting is constrained and only Z translation is permitted. The loading is applied in a similar manner to the previous experiment, but the simulation time is increased to around 500 ms. The level of hourglass energy for the inner tube is monitored during the analysis to ensure that it was within acceptable limits as defined by the LS-Dyna developers [20]. A number of simulations with different global damping coefficient were carried out and it was found that the global damping coefficient of 0.15 produced satisfactory results. Moreover, selective mass scaling (SMS) was invoked to achieve a reasonable runtime. The kinetic energies were monitored throughout the analysis

to ensure that it was negligible compared to the total internal energy, as recommended by LS-Dyna developers for quasi-static simulations.

IV. DESIGN OF THE MUSCLE

A. Effect of varying end-fitting diameters

The braid structure experiences a range of diameter changes, along its length, due to the axial displacement of one of the end-fittings. The part of the braid attached to the end-fitting experiences no change in diameter and restricts the diameter change of the sleeve thus maximum radial deformation can be seen in the mid-section, as shown in Fig. 3(a). Therefore, the commercially available braid structure, that is to be used for the physical prototype, can be annealed at different diameters to achieve the cylindrical form for the braid. A brief analysis is performed, in order to study the effect of varying the end-fitting diameter on the mechanical characteristics of the muscle. Three different braid models were developed with varying end-fitting diameters and braid angles. Using these braid structures, six simplified muscle models with similar lengths in both states were produced, as shown in Fig. 4(a). The number of fibers in the braided sleeve were further reduced to 36 in order to minimize the computation time without affecting the coherence of the results. For the extensor state, the inner chamber is modeled as a simple hollow cylinder with a length of 50 mm. Both contractor and extensor states were analyzed to measure the blocked force and free displacement for each muscle. The simulation results are shown in Fig. 4(b). As the end-fitting diameter is increased, the blocked force and the free displacement decrease for the contractor state. On the contrary, for the extensor state, the blocked force and the free displacement increase. It can be seen that muscle-2 produces comparatively good forces and displacement in both states. The physical prototype is therefore manufactured with a braid structure having dimensions similar to muscle-2.

B. Effect of varying inner chamber length (Extensor state)

In order to check the effect of the inner chamber length on the performance of the actuator (extensor state), two different models were created. In the first model, the inner chamber is modeled as a simple hollow cylinder, while in the second model the inner chamber is modeled as a bellows, as shown in Fig. 4(c). Both inner chambers have the same thickness. The bellows-like profile helps in accommodating an inner chamber whose length is greater than the extensor state of the actuator. The blocked force and free displacement is simulated for both models by applying an actuation pressure of 0.2, 0.5, 0.8 and 1 bar. The results are shown in Fig. 4(c). It can be seen that for the same actuation pressures, the model with the inner chamber having a bellows profile is able to produce higher forces and free displacement in comparison to the model with the simple cylindrical chamber.

C. Physical Prototype of the Actuator

The physical prototype and its sectioned CAD rendering are

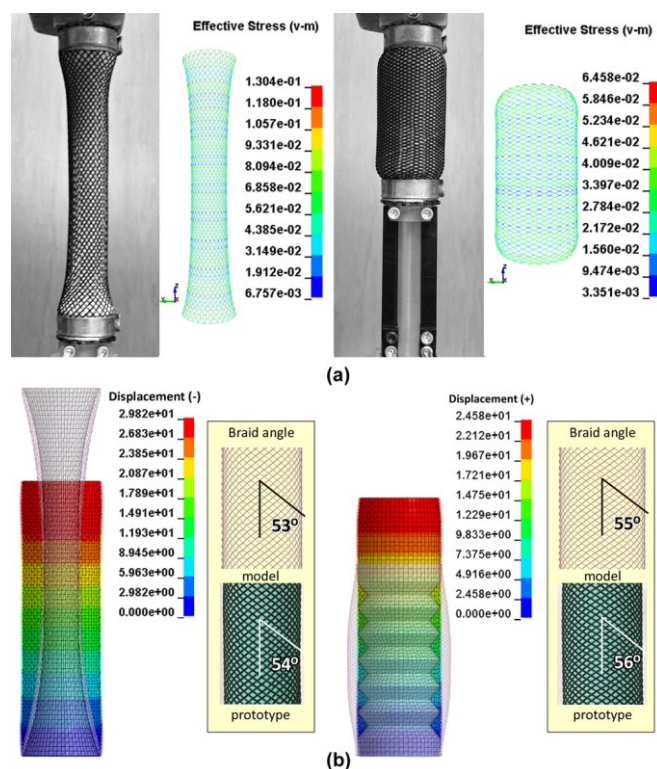


Fig. 5. (a) A comparison of the FE model and physical counterpart for the blocked force experiment with an actuation pressure of 1 bar. The von-mises stresses developed in the beam elements of the braid structure are displayed. Units: GPa. (b) Free displacement simulation results for both states, with an actuation pressure of 1 bar. The initial and deformed form of the contractor state and extensor state of the actuator model are shown along with a comparison of the braid angle for the actuated state of the model and the physical prototype.

shown in Fig. 1(b)(c). The end-fittings and the stem for the muscle were manufactured in house by utilizing rapid prototyping technology. Commercially available PET braid sleeving is used for the braid structure. It consists of 112 fibers arranged in an anti-symmetrical helical configuration in dual strands. The inner tube chamber is made of Latex with an approximate thickness of $0.25 \text{ mm} \pm 0.05$ and length of around $70 \text{ mm} \pm 2$. The outer diameter of each end fitting is 25 mm and the active length of the muscle ranges from 55 mm to 115 mm ± 0.5 , depending on the position of the movable end fitting. The movable end fitting can slide over the stem, hence enabling the user to change the braid fiber angle θ . For the extensor state, the braid angle is approximately equal to 71° while for the contractor state it is approximately equal to 28° at the center of the braid. O-rings are used to provide a sealing medium between the matting surfaces of the movable end fitting and the stem. The movable end-fitting can be displaced up to 60 mm over the stem. A passage is provided within the stem for pressurizing the inner tube. Another linear actuator could be used to automate the process, but for the current prototype, the end-fitting is manually moved and fixed at a certain position. The developed prototype is used to validate the FE models and also as the main driver for operating a one degree of freedom (DOF) variable stiffness joint, further discussed in section VII.

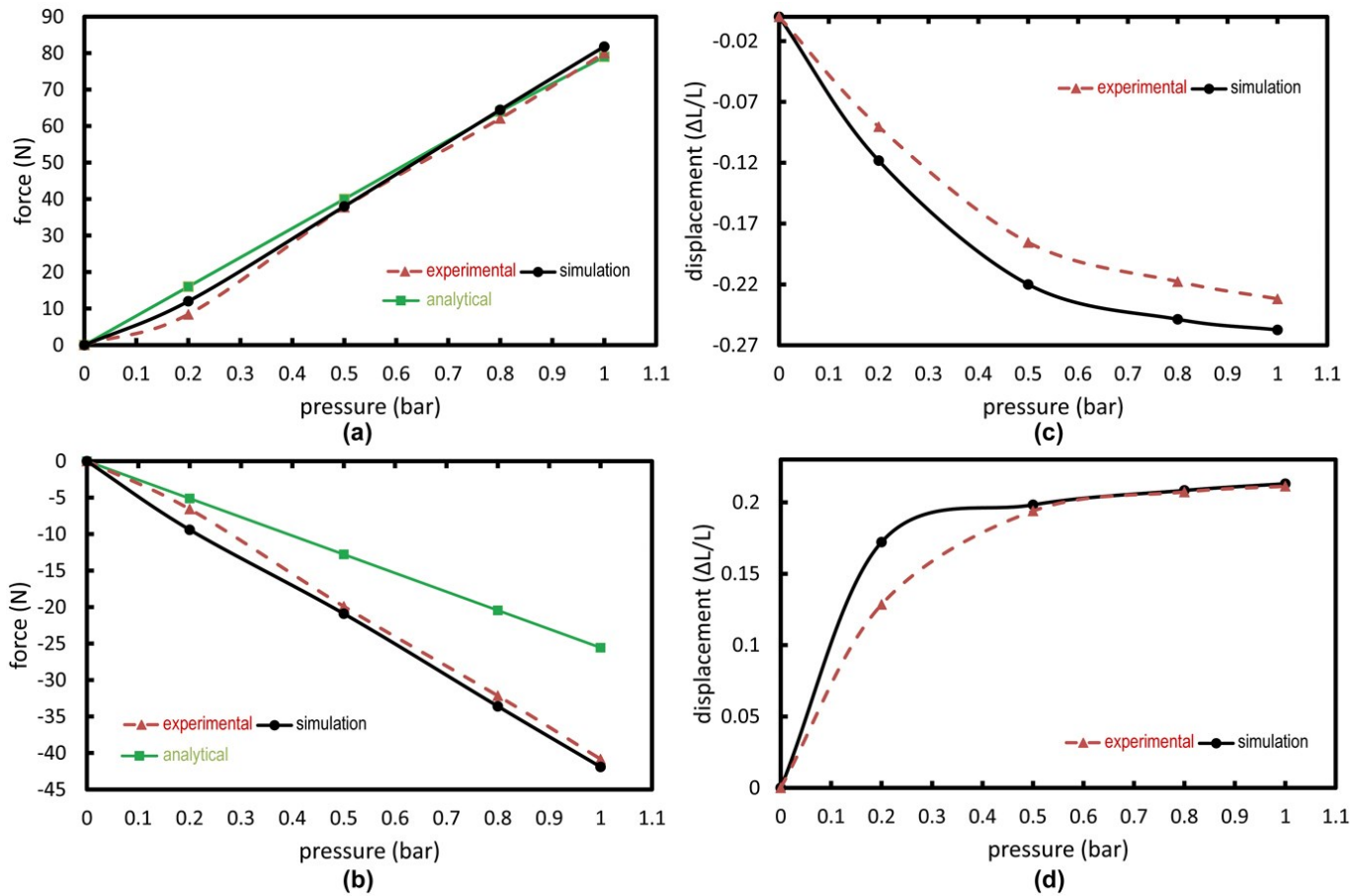


Fig. 6. Actuation pressure vs. applied load. Comparison of simulation and experimental results for the contractor state (a) and extensor state (b). For the blocked force the results of the analytical model [9] are also presented. Actuation pressure vs. free displacement. Comparison of simulation and experimental results for the contractor state (c) and extensor state (d).

V. MECHANICAL CHARACTERIZATION

The experiments were conducted in-order to characterize the bi-directional force output capacity, displacement and the axial stiffness of the actuator, under a range of internal pressure. The experimental setup consisted of a tension/compression testing machine (Instron 4464), air compressor and a pressure valve. An Arduino[®] Uno board is used to control the proportional valve, while a PC is used to collect data from the testing machine. First in order to validate the FE model two experiments were conducted on the muscle.

1) *Quasi-static Isometric tests*: Both end-fittings of the actuator are clamped in the testing machine. While keeping the length constant, the muscle is pressurized to predetermined pressure values of 0.2, 0.5, 0.8, 1 bar and the load values experienced by the machine are recorded.

2) *Quasi-static Isotonic tests*: Both end-fittings of the actuator are clamped in the testing machine. At the nominal position, the muscle is pressurized to predetermined pressure values of 0.2, 0.5, 0.8, 1 bar and kept constant. The top end-fitting is slowly moved until the load experienced by the testing machine approaches zero. The displacement of the top end-fitting is obtained.

In the last set of experiments, the axial stiffness of the physical prototype is measured for both states. The actuator is

pressurized and allowed to expand or contract freely. The top end-fitting is displaced 5 mm, either compressing or extending the actuator depending on the state it is in. The required amount of force is recorded from the testing machine and the axial stiffness is calculated. Each experiment is repeated three times and the average value is reported. The maximum actuation pressure is kept at 1 bar due to thin nature of the latex internal tube. As it was earlier stated the braid structure requires a minimum force for the deformation (i.e. less than 1 N). During the extensor state, the inner chamber is wrinkled as its initial length is greater than 55 mm. For the contractor state, the chamber is stretched axially in order to increase its length. For simplifying the FE models, in the braid structure and the inner rubber chamber. The same assumptions were also made for the experimentation i.e. the stresses developed by the braided sleeve and inner chamber were ignored.

VI. RESULTS & DISCUSSION

A comparison of the physical prototype and FE model for the blocked force measurement, with an actuation pressure of 1 bar is shown in Fig. 5(a). The Von Mises stress in the braid structure, for both states, are within acceptable limits. Similarly, the simulation results of free displacement for both

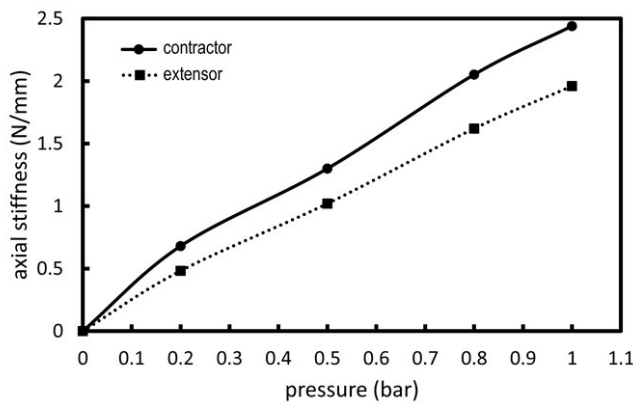


Fig. 7. Actuation pressure vs. axial stiffness. Comparison of experimental results for the contractor and extensor state.

states, with an actuation pressure of 1 bar is shown in Fig. 5(b). The measured blocked force and free displacement for the fabricated prototype are compared against quasi-static simulation results in an attempt to validate the FE model. The results for both the contractor and extensor states and there comparison is presented in Fig. 6.

The sign convention used is as follows: contraction force is denoted by a positive value and contraction displacement is denoted by a negative value, while extension force is represented by a negative value and displacement by a positive value. The results for the quasi-static isometric tests (experimental and simulations) can be seen in Fig. 6(a) and Fig. 6(b). It can be seen that the actuator is able to produce good force output, both in contractor and extensor states, with maximum forces of around 80 N (contractor) and -40 N (extensor). Higher forces can be achieved by increasing the actuation pressure and by changing the material properties for the different components of the muscle. It was also noted that a minimum (threshold) pressure of around 0.1 bar is required to actuate the muscle. This can be due to the fact that the inner elastic chamber and the outer braid are not in contact initially. This threshold pressure is required so that the inner chamber expands and comes in contact with the outer braid in order to produce an axial force output. Some hysteresis was also noted during the loading and unloading cycle (not shown here). This is an inherited drawback of PAMs, mainly due to the elastic nature of the inner chamber, friction between the inner chamber and the outer braid, as well as friction between the individual fibers of the braid during the inflation and deflation of the actuator. A close match can be found between the contractor model and the experimental results for the blocked force measurements with some discrepancies, especially at lower pressure value. Moreover, the model seems to follow the same trend as the physical counterpart. The blocked force, for both the simulations and the experiments, shows an almost linear increase with pressure: a behavior generally associated to PAMs.

The results for the quasi-static isotonic tests are presented in Fig. 6(c) and Fig. 6(d). The vertical axis in both figures is a non-dimensional measure of displacement. It is the ratio between the change in length upon actuation divided by the maximum active length ($\Delta L/L$), where $L=115$ mm, for both the

contractor and extensor states. It is evident that the actuator is able to achieve bi-directional motion with a maximum displacement of around 23% for the contractor state and around 21% for extensor state, with a maximum actuation pressure of 1 bar. The simulation and experimental results, for both states, show a nonlinear relationship between the actuation pressure and the free displacement, which tends to level off as the actuation pressure increases. This can be due to the fact that as the pressure increases, in both states, the fibers of the braid tends to reorient themselves and begins to reach a physical constraint. The further increase of actuation pressure will only increase the stiffness of the muscle with no effect on its free displacement. For the contractor state, the model seems to over predict the free displacement of the actuator with an almost constant error over different pressure values, but both the model and the prototype follow a similar trend. The extensor model seems to predict the behaviour of the muscle more accurately at higher pressures, but a significant error can be seen at the lower pressure value of 0.2 bar. One reason for such a behaviour can be the bellows profile used for the inner chamber, which tends to extend more than a simple cylindrical profile. Fig. 7 shows the relation between the applied pressure and measured axial stiffness of the actuator. It is evident that the axial stiffness increases almost linearly with increase in pressure for both the contractor and extensor states. Almost similar trend for both states of the actuator can be seen, with a maximum value of 2.44 N/mm for the contractor state and 1.96 N/mm for the extensor state.

VII. CASE STUDY: 1 DOF VARIABLE STIFFNESS JOINT

Future robots are expected to be inherently safe such that interacting with them should not possess a higher injury risk to humans or other robots. One method of achieving this aim is to use variable stiffness actuators [23][24], with rigid links. Applications for such mechanisms includes but are not limited to industrial automation, rehabilitation robotics, exo-skeletons etc. PAMs have been a popular choice among other actuators, as the main driver for such mechanisms, thanks to their inherently compliant structure and good force-to-weight ratio. As discussed in the introduction section, in order to carry out the bidirectional rotational motion of a joint by using conventional PAMs, two parallel muscles are required to work antagonistically. A number of such mechanisms have been proposed in literature [25][26][27]. In this section a brief introduction of a 1-DOF variable stiffness joint concept is presented, that uses the novel bi-directional muscle as its main driver.

The setup, shown in Fig. 8 (a), consists of a 4-bar mechanism, a smart servo motor (Robotis-Dynamixel), air compressor, a pressure valve and a load cell (ATI® Mini45-E Transducer). A detailed view of the 4-bar mechanism is shown in Fig. 8 (b). It is able to convert the linear motion of the top end-fitting of the muscle into rotational motion of the servo shaft and vice versa. The smart servo has an absolute encoder for position control and the torque can be controlled by varying the input current. The aim of using the bi-directional muscle is two folds. First, it acts as the main driver for the mechanism

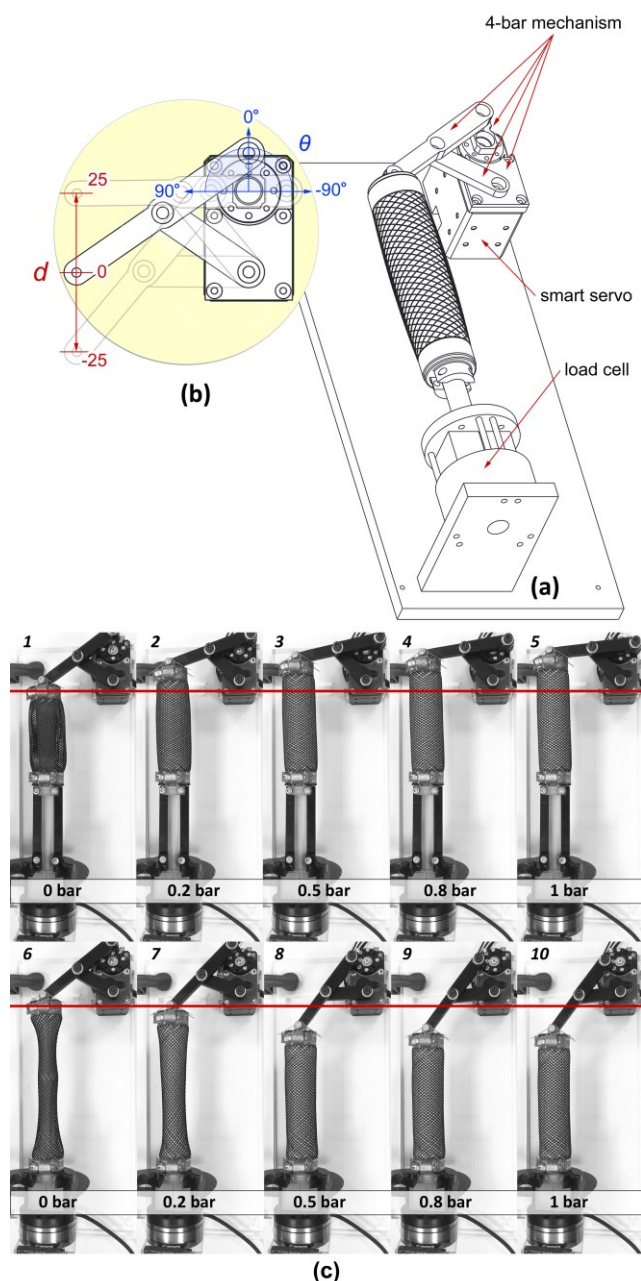


Fig. 8. (a) CAD rendering of the experimental setup. (b) Detail view of the 4-bar mechanism with maximum and minimum ranges for the linear displacement d and angular displacement θ . (c) Sequence of images showing the operation of the joint, where in the top sequence, the muscle is acting as an extensor while in the bottom sequence it is acting as a contractor.

and secondly it can act as a damper (energy dissipating medium) in case of an impact loading on the joint. The aim of using the smart servo is to maintain the top end-fitting at a certain position (0°), when the bottom end-fitting is shifted between the extensor and contractor state. This is achieved by initially applying a small torque to maintain the encoder position. When the bi-directional muscle is pressurized, the load produced by the muscle overcomes the torque applied by the motor and the joint moves. The operation of the joint is shown in Fig. 8 (d), where in the top sequence of images, the muscle is acting as an extensor while in the bottom sequence it is acting as a contractor. With an actuation pressure of 1 bar,

the muscle is able to produce a total linear stroke of around 46 mm enabling the joint to be rotated up to 160° ($\pm 80^\circ$). The difference in the force produced by the two states can be overcome by increasing the actuation pressure for the extensor state to achieve equal forces and response time.

VIII. CONCLUSION

In this study, the working concept and a physical prototype of the multi-functional pneumatic braided muscle actuator is presented as an improvement over the traditional pneumatic muscle actuator. The concept is first tested through a 3D-finite element model and then validated through experimental characterization. It is believed that the bi-directional actuation capabilities can notably enhance the abilities of PAM actuated robotic mechanisms/joints. For the last part, as a case study for the bi-directional actuator, a 1-DOF variable stiffness joint is briefly discussed. Some limitations are being investigated, including: (a) improving the FE model; (b) design of the linear actuator to actively control the displacement of the movable end-fitting; (c) developing a bellows-type, foldable inner tube, which can be easily deformed by the movable end-fitting and (d) further characterization of the variable stiffness joint. Such mechanisms can be used as a compact and powerful, yet inherently safer actuation approach for robotic systems and industrial automation.

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